Refbank: an open repository of iterated reference games

Veronica Boyce[1,2], Alvin W. M. Tan[1], Jess Mankewitz[1,3], Fiona Feng[4], Benji Fernandez[1], Ryder Fried[1], Ben Prystawski[1], Michael C. Frank[1], Robert D. Hawkins[1]
[1] Stanford, [2] MIT, [3] UW Madison, [4] Queen's University

Iterated reference games (IRGs) are a commonly used paradigm for studying adaptive language use, where people communicate about a set of potential referents. On each trial, one person, the *describer*, has a specific designated target that they seek to communicate to the *matcher(s)*. Across trials, targets are repeated so that the conversational history shapes the construction and understanding of later utterances (Fig 1).

IRGs are widely used to study convention formation, with each experiment manipulating a small set of possible experimental design variables and investigating a fraction of the possible analyses. Nevertheless, the underlying language data (and associated metadata) from IRGs constitute a high value corpus for studying not only convention formation, but also audience design, metaphor, semantic and syntactic alignment, and referring expressions; they can also be used as a benchmark for NLP models of pragmatics. **To enable this reuse, we present Refbank, an open repository of iterated reference game data.**

Refbank: Refbank stores data from IRGs in a relational schema, including the utterances, matcher's selections, and metadata about experimental conditions, the images, and participant demographics. A data processing pipeline creates derived measures, including semantic embeddings of utterances and similarity metrics across utterances (using S-BERT, Reimers and Gurevych, 2019). Refbank data is versioned and can be accessed through `refbank.github.io` or via a custom R package `refbankr`. The data can also be explored via interactive visualizations. Refbank currently has data from 13 datasets (12 English, 1 Dutch), including 1.3 million words from 127K referential trials from 2415 games and 7600 participants (Table 1). 7 datasets were online experiments where participants typed written messages, and 6 datasets were in-person experiments where participants spoke face-to-face. Studies varied in participant population, group size or partner switching structure, type and number of targets, and type of backchannel permitted. We are currently adding 4 more datasets with spoken interaction (2 English, 1 Swiss German, 1 French), comprising 9,000 trials (150 games).

Analyses: One goal of Refbank is to enable the study of moderators and generalizations of phenomena observed in IRGs. We provide three examples of how Refbank can replicate and extend prior findings. First, the canonical finding that the length of describer utterances decreases over repetitions replicates in Refbank, and Refbank allows exploration of how this trend is moderated by both group size and the modality of communication (Fig 2). Second, Hawkins et al. (2020) and Boyce et al. (2024) found that descriptions from adjacent repetitions become more similar in later repetitions. Refbank confirms that this observation holds across datasets and reveals that the convergence is stronger when the describer is constant across repetitions (Fig 3). Third, Hawkins et al. (2020) found that the distribution of parts of speech in describer utterances changes across repetitions, with nouns becoming relatively more common at the expense of closed-class words. In Refbank, this trade-off between nouns and closed-class words only occurs for datasets in the online written paradigm; in spoken experiments, nouns increase, but at the expense of verbs. We speculate that the ease of producing closed-class words may be modality-dependent, where it's easier to type telegraphic messages and easier to speak fluent descriptions with (phonologically reduced) closed-class items. This analysis illustrates how combining datasets allows for novel discoveries that can spur future research.

Future uses: Refbank can be used to study the moderators and generalizations of convention formation phenomena in IRGs, and identify gaps where future experiments could resolve theoretical questions. Refbank is also a corpus of task-specific dialogues that can be used for corpus linguistics, for assessing pragmatic phenomena in language models, and as a pedagogical resource for instructors and students alike. *We are seeking more datasets (including unpublished) to grow Refbank, so please contact us if you have an IRG dataset to contribute!*

Table 1: Included datasets

Refbank currently has language data from 13 datasets. Unless noted, each game had two adult players.

Paper	#Games	#Trials	#Words	Modality	Lang.	Notes
Branigan et al. (2016)	24	764	30 430	in-person	English	8-10yo children, partner swaps
Yoon & Brown-Schmidt (2019)	788	41 766	348 849	in-person	English	partner swaps, groups of 4-7
Leung et al. (2024)	82	3 259	37 360	in-person	English	parent &4-8yo child
Wang et al. (2025)	20	788	26 105	in-person	English	
Boyce et al. (2025)	97	1 438	16 887	in-person	English	4-5yo children
Bögels et al (2026)	47	4 512	176 483	in-person	Dutch	
Hawkins et al (2019)	168	3 994	19 740	online	English	
Hawkins et al. (2020)	100	7 120	62 590	online	English	
Hawkins et al (2021)	43	3 699	25 777	online	English	groups of 4 in changing pairs
Hawkins et al (2023)	26	2 278	11 756	online	English	groups of 4 in changing pairs
Eliav et al (2023)	262	8 183	50 468	online	English	
Boyce et al. (2024)	306	21 269	374 105	online	English	groups of 2-6
Mankewitz & Hawkins (2025)	447	28 244	198 713	online	English	
Total	2 415	127 401	1 380 037			

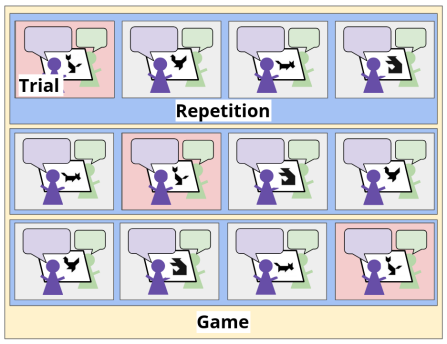

Figure 1: Schematic of IRG

Pink highlights show a target described in each repetition of an IRG.

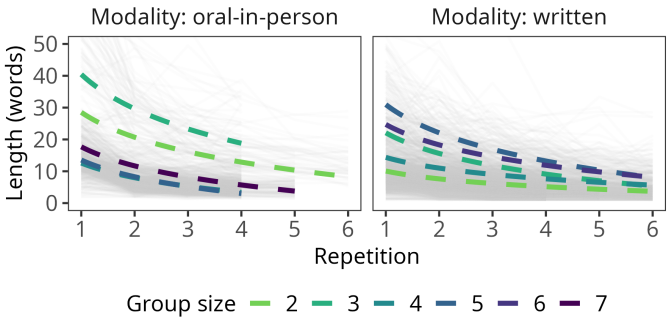

Figure 2: Reduction

Length of describer utterances across repetitions. Dashed lines are log best fit, faint lines are per game.

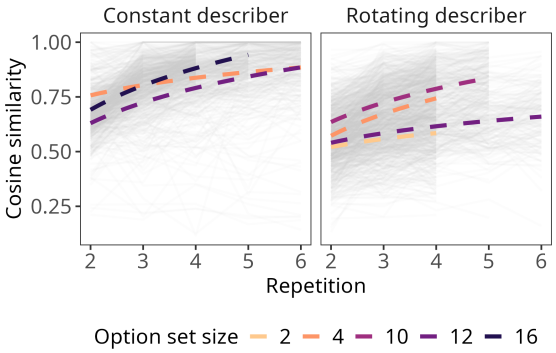

Figure 3: Convergence

Similarity of description and previous repetition's description of same target within games. Dashed lines are log best fit, faint lines are per game.

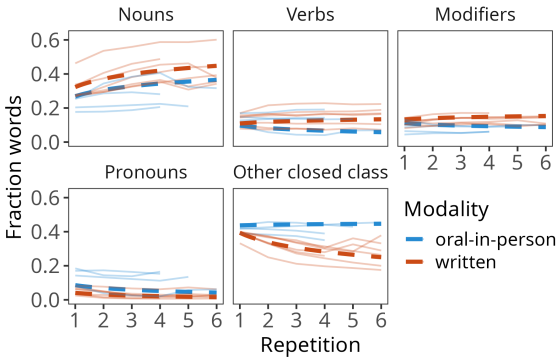

Figure 4: Parts of Speech

Fraction of words of describer utterances belonging to each category across repetitions. Dashed lines are aggregates, thin lines are per dataset.

Included datasets: Bögels, S., Li, T., et al. (2026). Cognition. doi.org/10.1016/j.cognition.2025.106370 • Boyce, V., Hawkins, R. D., Goodman, N. D., & Frank, M. C. (2024). PNAS. doi.org/10.1073/pnas.2403888121• Boyce, V., Sparks, R., Mofor, Y., & Frank, M. C. (2025). Cogsci Proceedings. escholarship.org/uc/item/047820tn • Branigan, H. P., Bell, J., & McLean, J. F. (2016). Frontiers in Psychology. doi.org/10.3389/fpsyg.2016.00213 • Eliav, R., Ji, A., Artzi, Y., & Hawkins, R. D. (2023). doi.org/10.48550/arXiv.2305.06539 • Hawkins, R. D., Kwon, M., Sadigh, D., & Goodman, N. D. (2019). CoNLL. doi.org/10.18653/v1/2020.conll-1.33 • Hawkins, R. D., Frank, M. C., & Goodman, N. D. (2020). Cognitive Science. doi.org/10.1111/cogs.12845 • Hawkins, R., Liu, I., Goldberg, A., & Griffiths, T. (2021). Cogsci Proceedings. escholarship.org/uc/item/6xz612cb • Hawkins, R. D., Franke, M., et al. (2023). Psychological Review. doi.org/10.1037/rev0000348 • Leung, A., Yurovsky, D., & Hawkins, R. D. (2025). Child Development. doi.org/10.1111/cdev.14186 • Mankewitz, J., & Hawkins, R. (2025). Cogsci Proceedings. escholarship.org/uc/item/4tw1c3gn • Wang, Z., Li, W., et al. (2025). EMNLP Proceedings https://aclanthology.org/2025.emnlp-main.849.pdf • Yoon, S. O., & Brown-Schmidt, S. (2019). Cognitive Science. doi.org/10.1111/cogs.12774

Other references: Reimers, N. and Gurevych, I. (2019). EMNLP arxiv.org/abs/1908.10084